

* L:9 (Prob. for ML)

* Revision:

- $n(A \cup B) = n(A) + n(B) - n(A \cap B)$
 - $n!$: number of ways to arrange n distinct objects.
 - n_{P_r} : " " " " " " " " out of n objects.
- $\left\{ \frac{n!}{(n-r)!} \right\} n_{P_r}$: Permutations

* n_{C_r} : Combinations

- Selection of object in which order does not matter.
- No. of ways to choose r objects out of n .

Ex: $\{A, B, C\}$

$\square \square$

$\left. \begin{array}{l} AB \\ AC \\ BA \\ BC \\ CA \\ CB \end{array} \right\} = \frac{3!}{1!} = 6$

$\{A, B, C\}$

$\downarrow$ combination of 2 chs.

$\{ -, - \}$

$\left. \begin{array}{l} AB \\ AC \\ BC \end{array} \right\} = 3 = \frac{3!}{2!}$

$$n_{C_r} = \frac{n_{P_r}}{r!} = \frac{n!}{(n-r)! r!}$$

* Probability:

Range: 0 to 1

It measures how likely an event.

0: never happens (sun rise in west) Impossible event

1: always " (" " " east) certain event

→ Random Experiment:

A process whose result can not be predicted with certainty, even though all possible results are known in advance.

Ex: toss a coin

→ Trial: one single performance of the random expe.

Ex: 100 times toss a coin: 100 trials

→ Sample space(s): The set of all possible outcomes of the expe.

Ex: toss a coin: $\{H, T\}$
die: $S = \{1, 2, 3, 4, 5, 6\}$

→ Event: Subset of sample space.

Ex: The die shows even number $\rightarrow \{2, 4, 6\}$

Fair Toss coin: $P(H) = \frac{1}{2} \rightarrow \frac{1}{2}$
→ $\{H\}$:

* How to calculate prob.:

$T/:$
 $\rightarrow$ S.S.

$$P(A) = \frac{n(A)}{n(S)}$$

$\downarrow$ Prob. of event A
 $\downarrow$ the favourable outcome
 $\downarrow$ no. of outcome in sample space.

$$\rightarrow \{2, 4, 6\} : \frac{3}{6} = \frac{1}{2}$$

Ex- die: $\{1, 2, 3, 4, 5, 6\}$ ← Sample space

What is the prob. of getting even number?

$$P = \frac{3}{6} = \frac{1}{2} = 0.5$$

$$P(A') = 1 - P(A)$$

①

* Complement Rule:

- Complement of A : (A') : A does not happen.

$$P(A') = 1 - P(A)$$

$$P(\text{sam. s.}) = 1$$



Probability of sample space = (?)

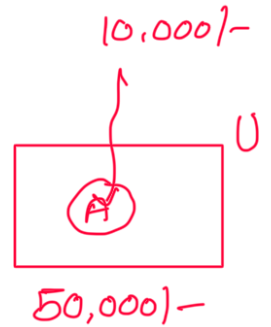
Roll a die :

$$S = \{1, 2, 3, 4, 5, 6\}$$

$$P(S) = (?) = \frac{n(S)}{n(S)} = 1$$

$n(\text{all balls})$
 $= 40,000$

$$n(U) = n(A) + n(A')$$



$$\frac{n(U)}{n(U)} = \frac{n(A)}{n(U)} + \frac{n(A')}{n(U)}$$

$$1 = P(A) + P(A')$$

$$P(A') = 1 - P(A)$$

$$* P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

* Conditional prob. : ←

$P(B|A)$: prob. of B given A

→ prob. that event B happens when we already know that event A has happened.

$$P(B|A) = \frac{n(A \cap B)}{n(A)}$$

$$P(A|B) \neq P(B|A)$$

$$\rightarrow P(A) = \frac{n(A)}{n(S)}$$

ABC



$$\rightarrow P(B|A) = \frac{2}{4} = \frac{1}{2}$$



P(having B at second place): $\frac{2}{6} = \frac{1}{3}$

P(having B at second place & A at first place): $\frac{1}{6} = \frac{n(A \cap B)}{n(S)}$

→ A is already at first place & it is fixed now

$\frac{ABC}{ACB} \approx \frac{1}{2}$

P(having B at second place)

A is already at first place & it is fixed now



Sample space

sample space
 $n(S) = 2$

P(having B at second place)

$\frac{n(\text{" " " " " "})}{n(S)} = 1$

$P = \frac{n(\text{event})}{n(S)} = \frac{1}{2}$

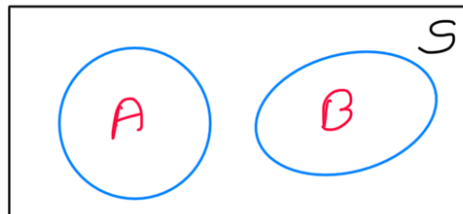
$$\begin{aligned}
 * \quad \underline{P(A \cap B)} &= \underline{P(A)} * \underline{P(B|A)} = \frac{n(A)}{n(S)} * \frac{n(A \cap B)}{n(A)} \\
 &= \underline{P(B)} * \underline{P(A|B)} = \frac{n(B)}{n(S)} * \frac{n(A \cap B)}{n(B)}
 \end{aligned}$$

- Special case: Independent event

* Types of event:

① Mutually exclusive event: (Disjoint)

- $A \cap B = \emptyset$
 - $\underline{P(A \cap B) = 0}$
 - $P(B|A) = 0$
- } Both can't be happen in the same trial.



② Independent event:

- If knowing that one occurred does not change the prob. of the other.

$$\underline{P(B|A)} = \underline{P(B)}$$

$$\underline{P(A \cap B)} = \underline{P(A)} * \underline{P(B)}$$

$$P(HHT) = P(H) * P(H) * P(T)$$

* Likelihood: Given particular parameter value, how likely is the data that we observed?

Q7

A coin: Not a fair coin } parameter
 $P(H) = 0.7$

Now we tossed that coin
 8 times Indepen...

We got HHT ←

$P(HHT | P(H)=0.7)$

→ then How can we calculate likelihood?

↳ the pro. of observing the data given particular parameter.

$$P(A|B) \rightarrow P(HHT | P(H)=0.7)$$

→ $P(HHT | P(H)=0.7) \Rightarrow$ three Independent toss

for inde. event $P(A \cap B) = P(A) * P(B)$

$$\begin{aligned} P(HHT | P(H)=0.7) &= P(H) * P(H) * P(T) \\ &= 0.7 * 0.7 * 0.3 \\ &= 0.147 \end{aligned}$$

→ what is the purpose of Likelihood?

$$P(HHT | P(H)=0.5) = (0.5)^3 \\ = 0.125$$

0.147 > 0.125 } the observe data HHT is more likely under $p=0.7$ than $p=0.5$

$$\begin{matrix} \downarrow & \downarrow \\ L(\theta | \text{data}) \end{matrix}$$

θ = Parameter

data = that we have already observed

L = Likelihood of θ

Likelihood of Parameter given observed data.

$$L(\theta | \text{data}) = L(P(H)=0.7 | HHT)$$

Parameter data

$$= P(HHT | P(H)=0.7)$$

$$= P(\text{data} | \theta)$$

→ $P(\text{data} | \theta) \approx P(\theta | \text{data})$:

Likelihood

pro. of observed data ^(Evidence)
given particular para.

Posterior

pro. of parameters
given the observed
data.

* Bayes' Theorem:

$$P(\theta | \text{data}) = \frac{P(\theta | \text{data})}{P(\text{data})}$$

$$= \frac{P(\theta) * P(\text{data} | \theta)}{P(\text{data})}$$



cat

?

dog

$$\rightarrow P(\text{cat}) = 0.5$$

$$P(\text{dog}) = 0.5$$

$$P(\text{cat} | \theta) =$$
$$P(\text{dog} | \theta) =$$

After observation
we get to know:

Pet is quit } New evidence

$$P(\theta | \text{cat}) = 0.8$$
$$P(\theta | \text{dog}) = 0.3$$

→ here we know our pet is quit but we want
proba. of dog & cat given it is quit.

$$P(\text{cat} | \theta) \quad \& \quad P(\text{dog} | \theta)$$

↓

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A) * P(B|A)}{P(B)}$$

$$P(\text{cat}|\text{g}) = \frac{P(\text{cat}) * P(\text{g}|\text{cat})}{P(\text{g})}$$

$$P(\text{cat}) \quad \checkmark$$

$$P(\text{g}|\text{cat}) \quad \checkmark$$

$$P(\text{g}) = ?$$

↓
Bayes theorem

$P(\text{g})$ = Prob. of quite animal (dog or cat any)

$$P(\text{g}|\text{cat}) = \frac{P(\text{g} \cap \text{cat})}{P(\text{cat})} * P(\text{cat})$$

$$P(\text{g}|\text{dog}) = \frac{P(\text{g} \cap \text{dog})}{P(\text{dog})} * P(\text{dog})$$

$$P(\text{g}) = P(\text{g} \cap \text{cat}) + P(\text{g} \cap \text{dog})$$

$$= 0.4 + 0.15 = 0.55$$

$$P(\text{cat}|\text{g}) = 72.7\%$$

$$P(\text{dog}|\text{g}) = 27.3\%$$

$$P(H|E) = \frac{P(E|H) * P(H)}{P(E)}$$

... (0.0.7)

H = Hypothesis $P(H)$
 E = data (HHT)
Observation

8.4 The four ingredients

Name	Symbol	Meaning in words
<u>Prior</u>	$P(H)$	What we believed about the hypothesis H before seeing any evidence.
<u>Likelihood</u>	$P(E H)$	How probable the observed evidence E would be if H were true.
<u>Evidence (marginal likelihood)</u>	$P(E)$	How probable the evidence is overall, across all hypotheses.
<u>Posterior</u>	$P(H E)$	Our updated belief about H after seeing the evidence. This is what we actually want.

$$P(p=0.7 | HHT) = \frac{P(HHT | p=0.7) * P(p=0.7)}{P(HHT)}$$

$$= \frac{\text{Likelihood} * \text{Prior}}{\text{Evidence}}$$

→ Baye's theorem helps to calculate posterior.

prob. (Q / observed data)

$$P(H|E) = \frac{P(H \cap E)}{P(E)} = \frac{P(H) * P(E|H)}{P(E)}$$